📝 Drafting Technical PRDs

Week 3 • Day 1 • Requirements Engineering & Mini-Capstone

TPM Academy

🚫 This Week is Non-AI

The discipline of structured thinking expressed as written words

AI fills vagueness with confident generic prose. Vagueness is the bug we're eliminating.

🎯 Day 1 Objectives

- Pick **one** feature concept from your Week-2 set, with explicit reasoning
- Write a **Context** section that gives an engineer a reason to care before reading what to build
- Write a **Problem** section grounded in your Week-1 interviews and Week-2 journey map
- State **goals + non-goals** as user outcomes, not feature names
- Write a **solution sketch** that lets an engineer imagine the shape without specifying the implementation

🕑 Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Pin Week-2 artifacts; restate the non-AI rule
09:15 – 10:30	Block 1 + Activity 1	Pick the one feature
10:45 – 12:00	Block 2 + Activity 2	Section 1 (Context) + Section 2 (Problem)
13:00 – 14:15	Block 3 + Activity 3	Section 3 (Goals/non-goals) + Section 4 (Scope)
14:30 – 16:00	Block 4 + Activity 4 + Wrap	Section 5 (Solution sketch)

Block 1 — What a TPM PRD Is

And how it differs from a generic PM PRD

📚 TPM PRD vs Generic PM PRD

Generic PM PRD	Technical PRD (your job)
Heavy on business framing	Engineer-readable from §1
Vague on edge cases	Granular AC + NFRs
Light on dependencies	Names systems, owners, gaps
"Out of scope" usually missing	Out-of-scope explicit
Risks listed as "low"	Real risks named honestly

The TPM contract: a PRD an engineering lead would accept as scoping input *without a clarifying call.*

📋 The 11-Section PRD Anatomy

This week's daily build

1. Context
2. Problem
3. Goals & non-goals
4. Scope (in / out)
5. Solution sketch
6. Acceptance Criteria *(Day 2)*

Continued

7. Non-Functional Reqs *(Day 3)*
8. Metrics & validation *(Day 4)*
9. Risks & open questions *(Day 4)*
10. Dependencies *(Day 4)*
11. Out-of-scope follow-ups *(Day 4)*

🎯 Pick One Feature, Defend the Choice

Your Week-2 set has 3 feature concepts. Today you pick one. Use the four-dimension scoring sheet:

Dimension	1	5
Friction severity	Mild annoyance	Task failure / lost data
Metric movability	Indirect, slow	Directly moves a Tier Sheet metric in 30 days
Build feasibility	>1 quarter	Achievable in 6–8 weeks
Learning value	Confirms what we know	Tests an unproven assumption

> **Tie-breaker:** if two features tie, the triad must *argue*, not split the difference. The discipline of choosing is the work.

👥 Activity 1 — Pick the One Feature

Triads • 35 minutes

Score your 3 Week-2 feature concepts on 4 dimensions. Pick one. Write a "Why this one" memo: 2 paragraphs, user terms + business terms.

- 5 min: re-read the 3 cards aloud
- 10 min: score each on 4 dimensions
- 10 min: pick + argue if tied
- 10 min: write the memo

⏱️ 5 read · 10 score · 10 pick · 10 memo

Block 2 — Context & Problem

Sections 1 and 2 of the PRD

📋 Section 1 — Context (½ page max)

What it answers

- Why this work?
- Why now?
- What changed?

What it avoids

- "Customers are asking for it" (vague)
- Marketing language
- Restating company strategy from scratch

The test: after reading §1, the engineer should be willing to keep reading §2.

📝 Section 2 — Problem (½ to ¾ page)

The user's **job-to-be-done**, the **friction they hit today**, why **workarounds are inadequate**.

Tie explicitly to the **Week-2 journey map**. Reference your friction stars by name.

❌ Generic

"Dispatchers struggle with reconciliation. The current process is inefficient and error-prone."

✅ Specific

"Dispatchers spend ~45 min after-shift cross-checking 3 systems (paper tickets, truck stock, timecards). Maria (Interview 2) said: 'I usually just

guess on the last 4 entries by 6:30 because I want to get home."'

👥 Activity 2 — §1 + §2

Triads • 40 minutes

Each member drafts §1 alone, then triad picks the strongest. Repeat for §2. Quote a real customer at least once.

- 10 min: solo §1 drafts
- 10 min: pick + combine
- 10 min: solo §2 drafts
- 10 min: pick + combine + quote check

⏱️ 10 + 10 + 10 + 10

Block 3 — Goals, Non-Goals, Scope

Sections 3 and 4 — bound the work

🎯 Section 3 — Goals (3–5)

- Each goal is a **user outcome**, not a feature
- Each goal is **observable within 30 days** of ship
- Each goal ties to a **Tier Sheet metric** from Week 2

❌ Feature-shaped

"Ship the new reconcile dashboard"

✅ Outcome-shaped

"Dispatchers see open work without hunting; median time-to-close drops 30%."

🚫 Non-Goals (2–4) — the under-valued section

Non-goals state what you **explicitly will not do**. Done well, this section pre-empts most scope-creep negotiations later.

FieldPulse examples:

- "We will not redesign the navigation bar"
- "We will not change the underlying data model for tickets"
- "This iteration will not support tablets"

Coach's prompt: "What's the most ambitious thing a stakeholder might think this includes?" Put that in non-goals.

📍 Section 4 — Scope (in / out)

In scope	Out of scope (this iteration)
Mobile flow	Tablet layout
Online + offline draft	Offline-first sync
English only	Spanish localization
Single-shop dispatchers	Multi-shop / franchise admins

The right column is the negotiation tool: "I see what you might also want; here's why this iteration doesn't include it."

👥 Activity 3 — §3 + §4

Triads • 40 minutes

Draft 5 candidate goals; cull to 3. Draft 3 non-goals; cull to 2. Build the in/out scope table.

- 10 min: goals (draft 5, cull to 3)
- 10 min: non-goals (draft 3, cull to 2)
- 15 min: scope table
- 5 min: sanity-check goal ↔ metric ties

⏱️ 10 + 10 + 15 + 5

Block 4 — Solution Sketch

Just enough for an engineer to imagine the shape

✏️ What Goes In §5

✅ Include

- User-visible flow (4–8 steps)
- Key surfaces affected
- Hard interactions with other systems
- Happy-path narrative (one paragraph)
- Optional: 1–2 paper sketches

❌ Exclude

- Database schema decisions
- Specific API contracts
- Class names, library picks
- Pixel-level layout

❓ The "Engineer's First Three Questions" Test

After writing §5, ask: what are the **first three questions** an engineer would ask after reading this?

Their first question is about	Diagnosis
Implementation choice	You over-specified; pull back
User behavior or scope	You under-specified; add detail
Edge cases / error states	That's correct — AC tomorrow will address

👥 Activity 4 — §5 Solution Sketch

Triads • 45 minutes + Wrap

Sketch the flow on paper. Write the happy-path paragraph. List hard interactions. Run the three-questions test.

- 15 min: sketch flow (4–8 steps)
- 10 min: happy-path paragraph
- 10 min: hard interactions list
- 10 min: three-questions test

⏱️ 15 + 10 + 10 + 10 · 15-min wrap-share

🎉 Day 1 Wrap

What we built

- Feature locked in with reasoning
- §1 Context (engineer-readable)
- §2 Problem (with customer quote)
- §3 Goals + non-goals (user outcomes)
- §4 Scope (in/out table)
- §5 Solution sketch (no over-spec)

What opens tomorrow

- **Granular Acceptance Criteria**
- Given/When/Then form
- The 5 AC failure modes
- Peer-AC review

Bring to Day 2: Your draft §§1–5. AC will be written against them.

💬 Questions & Reflection

If an engineer read your §5 right now, what would their first question be?